Propranolol Normalizes Metabolomic Signatures Thereby Improving Outcomes After Burn

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Objective and Background: Propranolol, a nonselective beta-receptor blocker, improves outcomes of severely burned patients. While the clinical and physiological benefits of beta-blockade are well characterized, the underlying metabolic mechanisms are less well defined. We hypothesized that propranolol improves outcomes after burn injury by profoundly modulating metabolic pathways.

Methods: In this phase II randomized controlled trial, patients with burns ≥20% of total body surface area were randomly assigned to control or propranolol (dose given to decrease heart rate <100 bpm). Outcomes included clinical markers, inflammatory and lipidomic profiles, untargeted metabolomics, and molecular pathways.

Results: Fifty-two severely burned patients were enrolled in this trial (propranolol, n = 23 and controls, n = 29). There were no significant differences in demographics or injury severity between groups. Metabolomic pathway analyses of the adipose tissue showed that propranolol substantially alters several essential metabolic pathways involved in energy and nucleotide metabolism, as well as catecholamine degradation ($P < 0.05$). Lipidomic analysis revealed that propranolol-treated patients had lower levels of proinflammatory palmitic acid ($P < 0.05$) and saturated fatty acids ($P < 0.05$) with an increased ratio of polyunsaturated fatty acids ($P < 0.05$), thus shifting the lipidomic profile towards an anti-inflammatory phenotype after burn ($P < 0.05$). These metabolic effects were mediated by decreased activation of hormone-sensitive lipase at serine 660 ($P < 0.05$) and significantly reduced endoplasmic reticulum stress by decreasing phospho-JNK ($P < 0.05$).

Conclusion: Propranolol's ability to mitigate pathophysiological changes to essential metabolic pathways results in significantly improved stress responses.

Key words: burns, hypermetabolism, lipolysis, propranolol

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Burns are among the most devasting forms of injury, accounting for a global estimate of 11 million annual cases requiring medical attention, of which over 180,000 patients succumb to burn injury.[1] Severe burns covering over 20% of total body surface area elicit an unparalleled hypermetabolic response that leads to generalized catabolism associated with substantial morbidity.[2] Hypermetabolism is thought to be mediated by an increase in endogenous catecholamines.[3] Indeed, burn-induced stress activates the sympathetic nervous system to release catecholamines resulting in a cascade of metabolic and inflammatory processes that alter metabolism for years.[4]

While severe burns affect every organ in the body, there is growing evidence that adipose tissue plays a crucial role in regulating and modulating postburn hypermetabolism. Studies have demonstrated that white adipose tissue undergoes a substantial remodeling process involving increases in the endoplasmic reticulum (ER) stress,[5] lipolysis, and "browning,"[6–8] whereby white adipocytes convert into an energy-expending state phenotypically similar to a brown adipocyte.[9] It is suggested that these factors perpetuate hypermetabolism by continuously fueling the catabolic fire.[10]

Since catecholamines released via sympathetic activation mediate hypermetabolism, propranolol, a nonselective β_1/β_2 blocker, has been proposed to mitigate burn-induced systemic consequences.[11] Propranolol is both safe and effective in attenuating hypermetabolism in children with severe burns.[12,13] Herndon et al[14] showed that long-term administration of propranolol in severely burned children resulted in a 110% and 20% decrease in heart rate and resting energy expenditure (REE), respectively. Patients also exhibited a 15% reduction in the accrual of central mass at 3 months. Moreover, Barrow et al[15] demonstrated that propranolol reduces the expression of genes related to lipid accumulation in pediatric burn patients. A recent study found that propranolol is safe and effective in severely burned adults, resulting in diminished muscle wasting and enhanced wound healing.[16] While the favorable effects of propranolol on the pathophysiology of a burn patient are delineated, the mechanisms by which propranolol reduces hypermetabolism are poorly understood. We, therefore, conducted a phase II randomized controlled trial to provide mechanistic insight towards understanding propranolol's beneficial effects in burn patients. Specifically, we hypothesized that adipose tissue is the central mediator of hypermetabolism, and the use of propranolol attenuates burn-induced metabolic dysfunction by targeting this dynamic endocrine organ.

METHODS

Study Design and Participants

Patients with an acute burn injury admitted to our provincial burn center between January 2011 and December 2020 were included in this cohort study. The research ethics board approved the study protocol (Study #373-2010b). Inclusion and

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exclusion criteria are outlined in Supplemental Digital Content 1, Table 1 (http://links.lww.com/SLA/E706). Patients were assessed for eligibility to be enrolled in the study by a research team member. After obtaining informed consent, the patients were randomized to the 2 groups using random block sizes to achieve balance in the number of patients assigned to each group while maintaining allocation concealment (Supplemental Digital Content 2, Figure 1, http://links.lww.com/SLA/E707).

The study protocol was initiated as soon as possible after entry into the study. Patients were treated until discharge from the hospital. Propranolol dosing was started at 10 mg 4 times a day and increased as needed to target a heart rate <100 bpm. Propranolol was decreased or held if bradycardia (< 50 bpm) or mean arterial pressure (MAP) fell <55 mm Hg. Propranolol was then restarted at half the dose when it was stopped. The dose was adjusted according to a scale by our team of intensive care nurses assisted by a study physician and the physicians involved in the clinical care of the patients.

On admission, all patients underwent standard burn care (eg, resuscitation, surgeries, nutrition, infection control, inhalation injury, etc.). All patients received continuous feeding starting within 12 hours after admission by either nasogastric, nasoduodenal/nasojejunal, or by mouth. We feed our patients a high-protein, high-glucose, low-fat diet (Isosource HN); the feeding rate is based on REE measurements and calculated by the Toronto prediction. We add a stress factor of 1.4 to our calculation.

All patients followed the study protocol. An oral glucose tolerance test (OGTT) was conducted at discharge when the patient was 95% healed.

Biospecimen collection and processing: Blood was drawn at admission and twice weekly until discharge. Blood samples were processed and frozen at −80°C until analysis. These samples were used for biomarker measurements, cytokine profiling, and acute phase protein analysis. Whole adipose tissue was collected at the time of surgery and immediately sent to the laboratory, where ~500 mg of tissue was placed in a cryogenic tube and frozen at −80°C.

Clinical Outcomes

Patient demographics and injury characteristics were collected. Patient clinical outcomes were recorded prospectively during rounds and entered into a database. Outcomes included length of stay, mortality, and in-hospital complications such as infections, pneumonia, and septic episodes defined by the American Burn Association guidelines.[17]

Clinical biomarkers measured were troponin T, creatine kinase, creatinine, lactate, pH, blood urea nitrogen, alkaline phosphatase, alanine aminotransferase, and bilirubin.

Inflammatory Responses

Inflammatory responses were determined in serum; cytokine profiling was conducted twice weekly until discharge by the Bio-Plex Suspension Array System.

METABOLIC RESPONSES

Glucose

Blood glucose was continuously measured during hospitalization. Data are presented as daily average, 6 AM, daily minimum, and daily maximum glucose levels.

We determined insulin sensitivity via OGTTs conducted when a patient was 95% healed. Standard OGTT consisted of ingesting 75 g of glucose and subsequent glucose, insulin, and C-peptide measurements were conducted over a period of 2 hours[18] and calculated according to well-established protocols. As previously described, we calculated insulin sensitivity indices: the quantitative insulin sensitivity check index, the composite whole-body insulin sensitivity index (composite), and the insulinogenic index.[18] Furthermore, REE was measured with a Sensor Medics 2900 metabolic measurement cart as previously described.[19,20]

Lipidomic Profile

Circulating free fatty acids (FFAs) from plasma were analyzed by gas chromatography-mass spectrometry performed by the Analytical Facility for Bioactive Molecules (AFBM) platform at the Hospital for Sick Children (Toronto, ON, Canada) as previously published.[21]

Tissue Metabolic Response

Protein lysates were prepared from adipose excised during operations occurring at > 10 days postburn. A RIPA lysis buffer was used, and extracts were separated on an SDS-PAGE gel before immunoblotting as described. Antibodies for binding immunoglobulin protein (BiP; 1:1000), phospho-JNK (p-JNK; 1:1000), β-actin (1:1000), phospho-sensitive and total hormone–sensitive lipase [p-HSL (Ser 660: 1:1000) and t-HSL (1:1000), respectively] were obtained from Cell Signaling Technology (MA) and incubated with the membrane overnight. Secondary antibodies consisted of horseradish peroxidase-conjugated anti-rabbit antibodies (1:3000) incubated for 1 hour. The membrane was washed in TBST twice for 5 minutes, and detection of the desired protein was achieved with a BioRad ChemiDoc Imaging System and SuperSignal West Pico Chemiluminescent Substrate (Thermo Scientific). Densitometry was performed using Image J for Windows.

Untargeted Metabolomics

Adipose tissue samples were shipped in dry ice to The Metabolomics Innovation Centre (TMIC), Edmonton, Alberta. Metabolites were extracted by homogenization with ceramic beads in precooled liquid-chromatography mass spectrometry–grade methanol:water. Chemical isotope labeling and metabolite identification were performed for each aliquot of the sample based on Tier 1 and Tier 2 criteria according to TMIC protocols (https://metabolomicscentre.ca/).

Quantitative Polymerase Chain Reaction (PCR)

Total RNA was isolated from adipose tissue and analyzed by quantitative reverse transcription PCR using TRIzol-chloroform (Life Technologies) with subsequent purification using the RNeasy Kit (Qiagen) according to the manufacturer's instructions. RNA (2 μg) was transcribed to complementary DNA using the high-capacity complementary DNA reverse transcription kit (Applied Biosystems). Real-time quantitative PCR was performed using the Applied Biosystems Step One Plus Real-Time PCR System.

Statistical Analysis

Data were reported in accordance with the Consolidated Standards of Reporting Trials (CONSORT) statement.[22] Continuous normally distributed variables were analyzed using the Student *t* test, analysis of variance, and continuous non-normally distributed variables were analyzed using the Mann-Whitney *U* test. Categorical data were analyzed using the Fisher exact test. Unless otherwise specified, statistical tests were 2-tailed, with a *P* value of <0.05 considered statistically

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TABLE 1. Demographics, Injury Characteristics, and Outcomes by Propranolol Administration

	n (%)		
	Control	Propranolol	P*
No. patients	29	23	
Age [median (IQR)] (y)	48 (36–59)	46 (34–58)	0.963
Male	21 (72)	19 (83)	0.513
Female	8 (28)	4 (17)	
TBSA [median (IQR)] (%)	42 (26–53)	40 (25–53)	0.789
Third TBSA [median (IQR)] (%)	25 (15–44)	24 (9–39)	0.606
Inhalation injury	15 (52)	9 (39)	0.412
BMI [median (IQR)] (kg/m^2)	26.0 (23.9–30.6)	23 (20.9–33.5)	0.484
Etiology			
Flame	26 (90)	17 (74)	0.161
Scald	1 (3)	2 (9)	0.577
Other	2 (7)	4 (17)	0.387
Complications			
Acute kidney injury	3 (10)	3 (13)	1.000
Sepsis	10 (35)	5 (22)	0.369
Pneumonia	15 (52)	13 (57)	0.785
Surgical site infection	7 (24)	6 (26)	1.000
Wound infection	7 (24)	3 (13)	0.482
LOS [median (IQR)] (d)†	38 (25–61)	40 (30–66)	0.605
LOS/TBSA [median (IQR)] (d/%)†	1.1 (0.8–1.3)	1.1 (0.8–1.6)	0.709
Mortality	3 (10)	1 (4)	0.621

Numbers may not add to 100 due to rounding.

*P values were calculated using the Mann-Whitney U test for continuous variables and Fisher exact test for categorical variables.

†Analysis restricted to patients alive until discharge.

BMI indicates body mass index; IQR, interquartile range; LOS, length of stay; TBSA, total body surface area.

significant. For the metabolomics, analyses and figures were generated using MetaboAnalyst (https://www.metaboanalyst.ca/). Otherwise, analyses were performed using SPSS Statistics, version 26.0. Figures were generated using GraphPad Prism, version 9.0.

RESULTS

Patient Demographics and Injury Characteristics

There were 52 patients included in this clinical trial (Supplemental Digital Content 2, Figure 1, http://links.lww.com/SLA/E707). Demographics, injury characteristics, and outcomes are shown in Table 1. There were no significant differences in ethnicity or preexisting medical conditions between the 2 groups (data not shown). Injury severity was similar between controls and propranolol (Table 1).

Clinical Outcomes

In terms of clinical outcome data, we found that the length of stay and length of stay normalized per percent burn were similar between groups (Table 1). As expected, due to low patient numbers, there was no significant difference between groups for the incidence of acute kidney injury, sepsis, pneumonia, surgical site infection, or wound infection (Table 1). Mortality was not different between propranolol and controls (Table 1 and Kaplan-Meyer Curve, Supplemental Digital Content 3, Figure 2, http://links.lww.com/SLA/E708).

Propranolol decreased heart rate as expected. We furthermore determined MAP and found no significant difference between groups. Propranolol was held in 13 patients, on average, 6 times per patient. The most common reason was in preparation for the OR.

Organ Function

Propranolol had no significant effect on troponin or creatine kinase after burn; similarly, we found no difference for creatinine or blood urea nitrogen between the 2 groups ($P > 0.05$; Supplemental Digital Content 4, Figure 3, http://links.lww.com/SLA/E709). We found no difference between groups for lactate and pH after burn ($P > 0.05$; Supplemental Digital Content 4, Figure 3, http://links.lww.com/SLA/E709). Propranolol had no effect on alkaline phosphatase, alanine aminotransferase, or bilirubin when compared with controls ($P > 0.05$; Supplemental Digital Content 4, Figure 3, http://links.lww.com/SLA/E709).

Inflammatory Response

In line with previous findings that interleukin (IL)-6 regulates adipose browning and the resulting increase in peripheral lipolysis, we analyzed serum from burn patients and found that propranolol significantly reduced IL-6 at the time period where adipose undergoes this phenotypic switch (Supplemental Digital Content 5, Figure 4A, http://links.lww.com/SLA/E710). However, no significant changes were observed with other proinflammatory mediators, such as IL-1β and tumor necrosis factor-α (Supplemental Digital Content 5, Figures 4B, C, http://links.lww.com/SLA/E710). This was also true of classically defined anti-inflammatory cytokines such as IL-4, IL-10, and IL-13, on which propranolol treatment had no effect (Supplemental Digital Content 5, Figures 4D–F, http://links.lww.com/SLA/E710).

METABOLIC RESPONSES

Glucose Metabolism

Mean, 6 AM, maximum, and minimum daily glucose levels were similar between groups (Supplemental Digital Content 6, Figure 5, http://links.lww.com/SLA/E711). However, propranolol-treated patients had significantly lower mean glucose levels and lower maximum levels 5 weeks postinjury ($P < 0.05$; Supplemental Digital Content 6, Figures 5A, C, http://links.lww.com/SLA/E711).

Results from OGTT, conducted at discharge, showed that propranolol improved insulin sensitivity during hospitalization ($P < 0.05$; Supplemental Digital Content 7, Table 2, http://links.lww.com/SLA/E712 and Supplemental Digital Content 8, Figure 6, http://links.lww.com/SLA/E713). Propranolol significantly reduced the amount of insulin and its byproduct C-peptide required to control glucose levels ($P < 0.05$). Propranolol restored the quantitative insulin sensitivity check index into the normal range, while control remained below normal indices (Supplemental Digital Content 7, Table 2, http://links.lww.com/SLA/E712). These results indicate that propranolol administration reduced insulin resistance, which agrees with previous studies.

Lipidomic Profile

Lipidomic analyses from the serum revealed that propranolol had a substantial effect on lipid metabolism. Propranolol administration was associated with significantly decreased levels of the inflammatory fatty acid palmitate at later time points following burn injury versus the group without treatment ($P < 0.05$; Fig. 1A). Moreover, total circulating nonesterified fatty acids were significantly decreased beyond week 1 ($P < 0.05$; Fig. 1B). When expressed as a percentage of the total lipidomic

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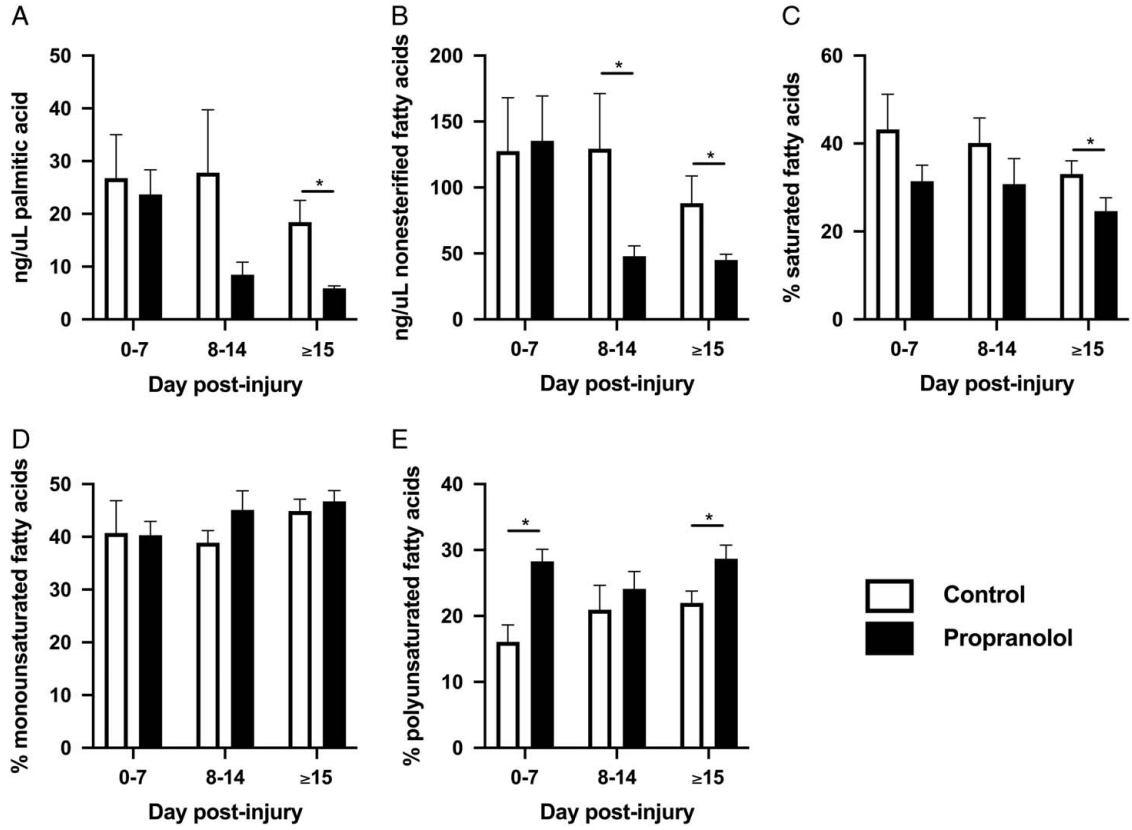


FIGURE 1. Lipidomic profile of burn patients by propranolol administration. Propranolol significantly decreased palmitic acid (A), nonesterified fatty acids (B), and saturated fatty acids (C). Propranolol had no effects on monounsaturated fatty acids (D), while it significantly increased polyunsaturated fatty acids (E). Error bars indicate SEM. *P* values were calculated with the Student *t* test. $^*P < 0.05$.

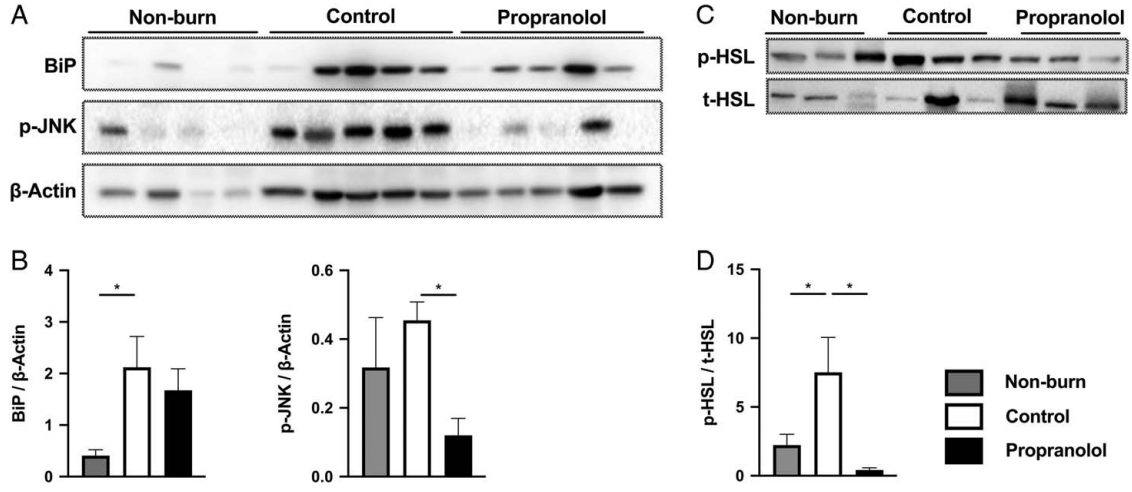


FIGURE 2. ER stress and lipolysis of burn patients by propranolol administration. Propranolol significantly reduced ER stress as shown by decreased binding immunoglobulin protein expression (A). Propranolol further decreased phosphorylation of the ER stress signaling marker JNK (B). Propranolol significantly decreased phosphorylation of HSL (C, D). Error bars indicate SEM. *P* values were calculated with the Student *t* test. $^*P < 0.05$.

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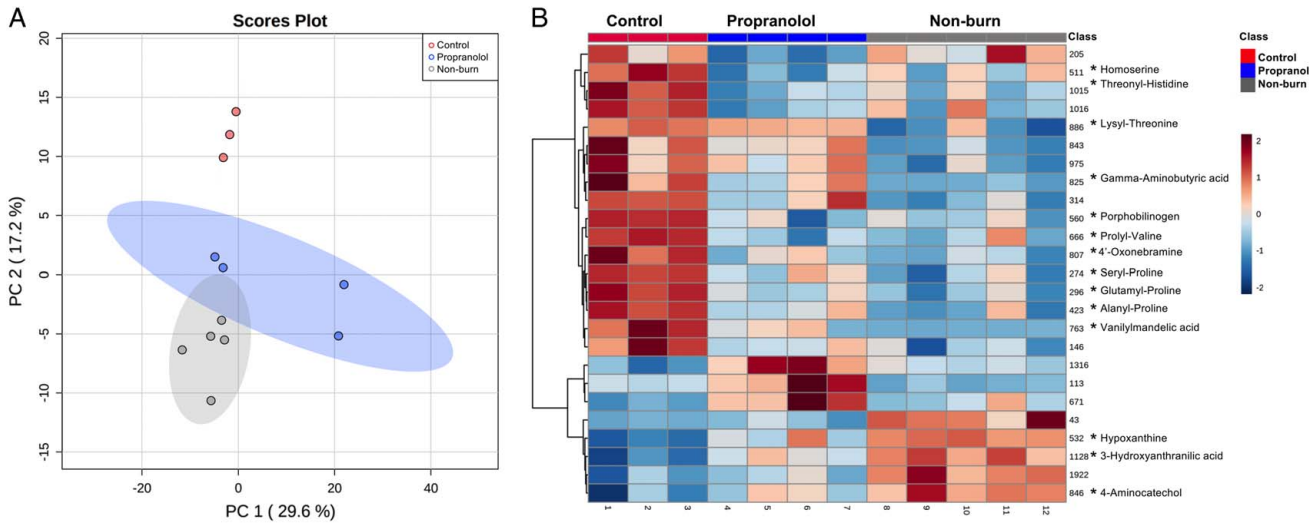


FIGURE 3. A, Principal component analysis (PCA) of metabolomic profile in the adipose tissue. PCA plot representing metabolomic datasets for the nonburn group, the control group, and the propranolol group. B, Heatmap of significantly altered metabolites in the adipose tissue. Heatmap representing concentration of the 25 most significantly altered metabolites between groups. Red color indicates high concentration of metabolite and blue color represents metabolites with low concentration. *$P < 0.05$. FC indicates fold change.

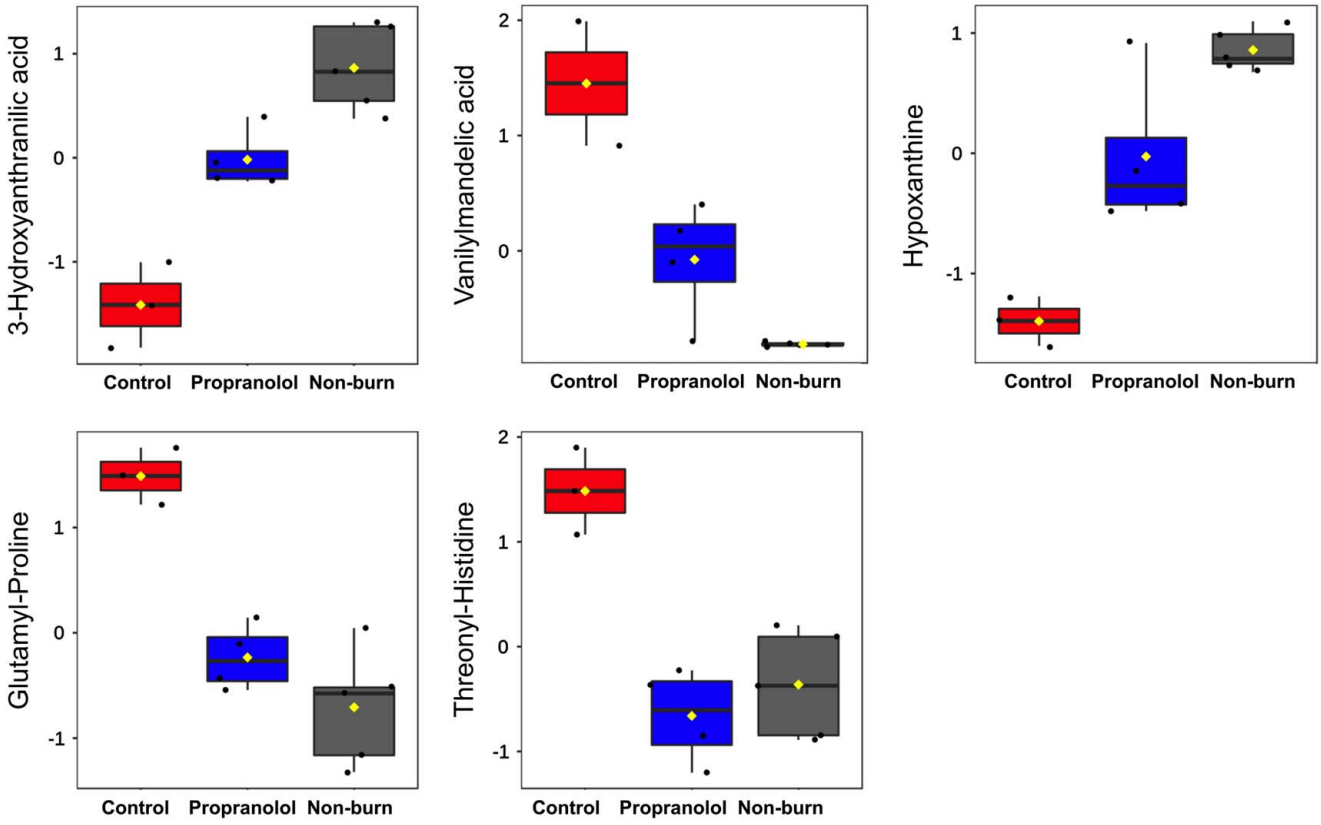


FIGURE 4. Propranolol restores metabolite concentration back to control levels in the adipose. Significantly altered metabolites (3HAA, VMA, hypoxanthine, glutamyl-proline, threonyl-Histidine). *P* values were calculated with a one-way analysis of variance. *$P < 0.05$.

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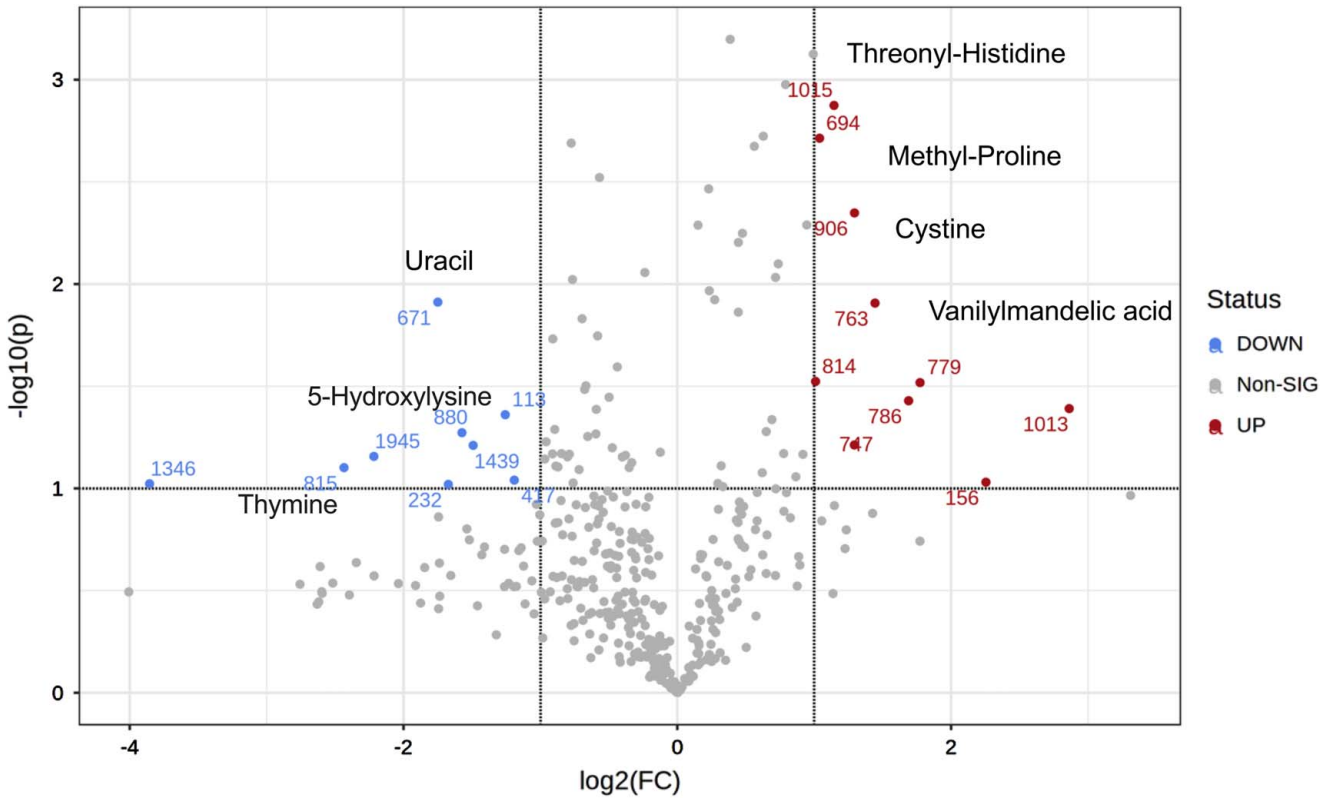


FIGURE 5. Volcano plot highlighting significantly altered metabolites in adipose tissue. Blue color indicates downregulated metabolites and red color indicates upregulated metabolites. Analysis refers to significantly altered concentrations between the control group and the propranolol group. *P* values were calculated with the Student *t* test. **P* <0.05.

profile, propranolol treatment in patients lowered the abundance of saturated fatty acids while increasing the proportion of polyunsaturated fatty acids in comparison to the control group ($P < 0.05$; Figs. 1C–E). Collectively, these results demonstrate that propranolol induces a shift in the release of circulating fatty acids.

Tissue Metabolic Responses

The alteration in the lipidomic profile prompted us to examine metabolic changes in the adipose tissue, where propranolol has previously been demonstrated to decrease peripheral lipolysis. ER stress markers, such as BiP and p-JNK, were analyzed, and propranolol significantly decreased the expression of the latter ($P < 0.05$; Figs. 2A, B). Moreover, as there is a strong link between ER stress and lipolysis, activation of HSL (Ser 660) was probed and shown to be reduced by beta-blockade ($P < 0.05$; Figs. 2C, D). Taken together, these data suggest that propranolol has antilipolytic properties in burn patients, thus decreasing the available fuel for thermogenesis.

Adipose Tissue Metabolic Profile

Due to the significance of adipose tissue changes after severe burns, we decided to investigate the metabolomic profile of the adipose tissue of propranolol-treated patients. We performed whole adipose tissue metabolomics, and principal component analysis showed a distinction between samples from burn patients and samples from propranolol-treated patients or nonburn patients (Fig. 3A). Untargeted metabolomic analysis revealed 18 significantly altered metabolites out of 404 identified compounds (Fig. 3B). Among these 18 metabolites, we observed that propranolol treatment normalized the concentration of 14 metabolites in burn patients to that of nonburn patients (Figs. 4, 5). These alterations brought to light the potential effects of propranolol in metabolic homeostasis that have not yet been studied. Most of the compounds identified are metabolites associated with inflammation, tissue remodeling, and mitochondrial health.

Moreover, pathway analysis revealed that propranolol treatment affects several metabolic pathways, such as energy metabolism (ie, arginine and proline metabolism, cysteine and methionine metabolism, beta-alanine metabolism, histidine metabolism, nicotinate and nicotinamide metabolism, tryptophan metabolism), nucleotide metabolism (ie, pyrimidine and purine metabolism), Vitamin B metabolism (ie, porphyrin and chlorophyll metabolism), and lipid metabolism (ie, sphingolipid metabolism, primary bile acid synthesis; Supplemental Digital Content 9, Table 3, http://links.lww.com/SLA/E714).

We isolated the normalized compounds and found 3-hydroxyanthranilic acid (3HAA), vanillylmandelic acid (VMA), hypoxanthine, glutamyl-proline, and threonyl-histidine to be the most significantly altered metabolites ($P < 0.001$) in burn patients, with levels normalized to nonburn patient levels after propranolol treatment (Fig. 4). Notably, 3HAA is a recently isolated metabolite that is part of the tryptophan and kynurenine metabolism pathway and one of the NAD^+ de novo biosynthesis

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pathways. 3HAA is an NAD^+ precursor that has been implicated in increased longevity, reactive oxygen species (ROS) regulation, and mitochondria health.[23] Propranolol treatment seems to restore 3HAA levels (Fig. 4). The next most significantly altered metabolite, VMA, is a byproduct of catecholamine degradation. This metabolite is increased in the adipose tissue of burn patients, which is expected as a means to deal with the intense catecholamine stress induced by severe burns. Propranolol attenuates the accumulation of VMA, bringing its levels down to similar concentrations as the nonburn patient group (Fig. 4). Hypoxanthine, glutamyl-proline, and threonyl-histidine are important metabolites in several processes, such as nucleotide synthesis, collagen formation, immune cell differentiation, and age-related adipose tissue dysfunctions.[24–32]

Finally, we found several significant metabolites that are either downregulated or upregulated in burn patients. These compounds are mostly involved in energy metabolism and nucleotide metabolism (threonyl-histidine, methionyl-proline, cystine, uracil, VMA, 5-hydroxylysine, thymine) (Figs. 4, 5). Histidine is a precursor for histamine that can also help scavenge ROS in inflammatory cells[24] and is upregulated in burn patients. Whereas 5-hydroxylysine, which is required for the formation and stability of triple helical collagen fibrils, is downregulated in burn patients compared with propranolol-treated patients.

DISCUSSION

In this prospective randomized controlled trial, we aimed to determine by which mechanisms propranolol affects postburn hypermetabolism. Even though the adipose tissue plays an essential role in the detrimental hypermetabolic response following burn injury, metabolite alterations in the tissue have been largely neglected. Here, we showed that propranolol treatment increases 3HAA concentrations to levels close to those of nonburn patients. 3HAA is a metabolite downstream of the tryptophan/kynurenine metabolism, which generates de novo NAD^+. Studies have shown that, in aged models, NAD^+ supplementation or NAD^+ precursors promote antioxidant and anti-inflammatory effects. They also show that 3HAA might act as a ROS scavenger and thus restores mitochondrial health.[23,33]

Another compelling finding was that VMA is significantly increased in burn patients, and propranolol treatment brings the levels down to those of the nonburn patient group. VMA is a metabolite downstream of the norepinephrine degradation pathway. It is mostly present in the periphery and highlights changes in catecholaminergic release.[34] As a catecholamine receptor blocker, we show that propranolol treatment might reduce VMA generation by inhibiting the synthesis and uptake of peripheral catecholamines, resulting in reduced demand for the adipose tissue to deal with the excess catecholamines and the downregulation of their degradation pathway.

Our pathway analysis revealed that a substantially altered pathway between burn patients and burn patients that received propranolol treatment was pyrimidine metabolism. This includes uracil and thymine metabolism, both compounds are downregulated in burn patients. Uracil can be generated via uridine conversion. In the liver, uridine supplementation suppressed lipid accumulation and increased the NAD^+/NADH ratio.[32] Moreover, even though uridine synthesis occurs mainly in the liver, during fasting periods, it is produced by adipocytes as the liver prioritizes glucose production.[35] Decreased uracil pools in the adipose tissue after a burn could be caused by pyrimidine deficiency or impaired metabolism. Another finding from metabolomics pathway analysis was the alteration in cysteine and methionine metabolism, being upregulated in burn patients that did not receive propranolol. Glutathione, a potent antioxidant, is synthesized from glutamate, cysteine, and glycine.[30] In contrast, high levels of cysteine can be toxic to mitochondria by limiting iron availability in the cytoplasm.[31,36] Cysteine restriction and iron supplementation have been shown to restore mitochondrial health during aging.[31] The fact that this metabolic pathway is increased in burn patients and downregulated after propranolol treatment shows that beta-blockade can impact mitochondrial health in adipose tissue via several unexplored mechanisms.

We showed that propranolol substantially decreased p-JNK, a key ER stress component and mediator of insulin resistance. We demonstrated that this decrease was associated with a reduction in adipose tissue lipolysis and decreased circulation of FFAs mediated by HSL. Lipidomic analysis further revealed that propranolol-treated patients had lower levels of proinflammatory serum palmitic acid and total saturated fatty acids in general with a circulating lipidomic profile that appears to shift towards an anti-inflammatory state with increases in the concentrations of polyunsaturated FFAs. We, therefore, suggest that propranolol affects the metabolic response after burn by blocking catecholamine-induced effects on the adipose tissue. Catecholamines drive vastly increased metabolic demands, and the energy requirements of these demands are met by mobilizing proteins and amino acids from peripheral tissues and burning these substrates as fuel.[37] As a consequence, the structure and function of essential organs and physiological processes such as the skeletal muscle, skin, immune system, and cellular membrane transport become compromised.[38,39]

In this trial, we used a propranolol dose that has been shown to decrease the heart rate to <100 bpm. Decreased heart rate, along with increased stroke volume, implies improved filling pressure, reduced cardiac work, and improved cardiac oxygen consumption.[13,40,41]

Cardiac dysfunction is known to be associated with serious burn injury, and it is one of the major contributors to morbidity and mortality.[42] In the present study, we have a strong indication that propranolol decreased heart rate with no difference in MAP, which results in beneficial effects in terms of cardiac work.

Glucose and fat kinetics are dramatically abnormal following burn injury.[43–46] Glucose utilization in burned patients is almost entirely through inefficient anaerobic mechanisms, characterized by increased lactate production, which accounts for increased glucose consumption.[3,47] A severe burn causes increased insulin levels,[46] which does not result in decreased glucose levels. In contrast, increased insulin is associated with increased serum glucose.[43] The fact that the basal rate of glucose production is elevated despite elevated plasma insulin levels indicates hepatic insulin resistance since, under normal conditions, elevated serum insulin would lower the rate of glucose production.[47,48] Hyperglycemia is associated with increased mortality in critically ill patients [49] and worsens the outcomes in severely burned patients.[50] In the present study, we showed that propranolol administration significantly improved insulin sensitivity. Patients receiving propranolol required dramatically less insulin and had lower blood glucose levels when compared with the control group, suggesting that insulin sensitivity was improved in the propranolol-treated group.

Insulin resistance is further linked to fat metabolism. Fat as an organ has gained attention due to its central role in the regulation of insulin resistance, mediating metabolism via browning, as well as many aspects of inflammation and immunity. Increased inflammation and lipolysis are potent inducers of fatty infiltration in the liver and other organs postburn. We have shown that

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hepatomegaly with fatty infiltration is associated with an increased incidence of sepsis and mortality, implying the importance of organ integrity and function in morbidity and mortality. In the present study, not only did propranolol improve insulin sensitivity, but it also decreased peripheral lipolysis, potentially reducing fatty infiltration of various organs.

We found that propranolol decreased inflammatory mediators after burn. IL-6 plays key roles in proinflammatory immunometabolic responses and has been associated with postburn thermogenesis.[51] Propranolol significantly reduced IL-6, which was associated with reduced proinflammatory lipid mediators as well. Propranolol attenuated proinflammatory saturated fatty acids that are associated with fatty infiltration of organs and induction of insulin resistance and cell organelle dysfunction, namely ER stress and mitochondrial dysfunction. Although we observed alterations in the adipose tissue, we did not show any difference in morbidity or mortality. The trial was not set up or powered for these outcomes.

There are 2 interesting findings in this trial. First, we found tremendous variability in biomarkers. We hypothesize that is due to the wide age range and differences in the age profile. This leads to suggest that newer interventional treatments should be personalized, and we should prospectively measure biomarkers and only intervene when there is an abnormality and a strong indication to treat.

In summary, we found that propranolol administration improves postburn metabolic and inflammatory responses by normalizing the metabolomic profile and restoring pathways affected by burn-induced stress.

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DISCUSSANT

Dr. Tina Palmieri (Sacramento, CA)

Tina Palmieri, Sacramento. Thank you for the privilege of this paper. I'd like to thank Dr. Jeschke for giving me a rough draft of the paper last week.

This paper extends Dr. Jeschke's previous extensive work with propranolol. Propranolol is like religion. Either you believe in it or you don't. This paper helps provide further data on mechanistic aspects of propranolol use. This phase II randomized clinical trial analyzed 52 patients gathered over a nine-year period and looked at a wide range of clinical factors, inflammatory markers, targeted and nontargeted metabolomics, as well as select markers of molecular activity and oral glucose tolerance tests.

Unfortunately, the study was underpowered to determine clinical efficacy of propranolol in the adult population. However, it does provide insights and hypotheses that we can further analyze in the future to further establish propranolol's effects on fat and fatty acid metabolism.

I have several questions. First, molecular and structural phenotypes of adipose tissue vary widely based on the adipose depot from which they are obtained. It also depends on when they were obtained and the circumstances under which they were obtained. Could you please provide us with further insight on the location of adipose tissue depots whether or not they were consistent between the patients, and the conditions under which the samples were obtained?

Second, of all your inflammatory markers, only IL-6 achieved significance. Why was there such variability, and why was it only IL-6? Can you hypothesize as to why that may be?

Third, given the differential impact of fatty acids on the inflammatory response, particularly in obese patients, could you provide us further insight into how frequently obesity occurred in each of your patient groups? Was it the same? You had twice the number of women in one group as you had in the other. Did the women have different responses?

And finally, how are we going to prove if propranolol actually after injury, during the treatment phase, is efficacious in adults? How would you design a trial to prove that propranolol actually does make an impact on mortality in the treatment phase in the hospital? Thank you again for a nice presentation.

Response From Marc Jeschke

Thank you, Dr. Palmieri, for your questions. The adipose tissue was collected at the time of surgery. We take the patients to their first OR within 72 hours or shortly thereafter. We published in Cell Reports that browning starts to occur at that time—Dr. Sidossis did that at the same time with Dr. Herndon in a study published in Cell Metabolism. We looked at when we take the adipose from various locations, is there a difference location-wise, and because it's peripheral fat and that undergoes browning, there is no difference, so we can safely assume regardless of where you debride it from the extremities, there should not be a difference. You will definitely find differences if you take the intra-abdominal fat or the brown fat per se which sits under the clavicle in humans, but we didn't study the brown fat. We just took white fat that is undergoing the browning response, and the timings were all pretty much the same, so we kept consistency there

I completely agree with you, the trial was not set up as an efficacy trial. It was really set up to answer molecular biological questions.

Why did we find only differences for IL-6 and why did we not find any difference for IL-1, IL-4, IL-13, because we showed this in pediatric patients before. We hypothesize that we don't see a significant difference because our population ranges over a wide age span and others and we believe that older adults and adults have a very different biological expression of these biomarkers, and we saw the variability. Adults definitely vary greatly in the expression of all these markers, which is very different to pediatrics patients. This large variability is the reason we cannot achieve significance with the current patient number.

You asked me about the obesity aspect. We looked at the BMI across these patients. There was no difference in terms of BMI. We can add this to the publication, so there was no difference in obesity, plus we showed and presented in Annals of Surgery, that actually patients who are slightly obese/slightly overweight have a better outcome than cachectic patients or massively obese patients, so there seems to be a protective effect of mild obesity.

And your last question, well, I think that Dr. Herndon tried to kick off this multicenter mortality-driven trial for propranolol, and I'm not sure where it is at this time. You may know better than I do where it is, but I didn't hear any updates, and I think the power calculation showed about 1000 patients needed for mortality differences. I do find that it's very challenging in burn patients to enroll 1000 patients. The first trial to do this was the recently p[published RE-ENERGIZE trial recently, and that took a long time in global enrollment, so it will be very difficult in my opinion to show a mortality benefit of propranolol.

And lastly as a comment, I agree with you. Propranolol administration is a little bit religion. Do I give propranolol to every one of my patients? No, I selectively pick my patients where I believe there is a benefit, and there are patients that I believe will not benefit from propranolol, so again, I'm agreeing with your comment as well.

Thank you very much for your questions.

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Dr. David Soybel (White River Junction, VT)

Thank you, very intriguing followup to many, many years of work. I had two questions. The first question was if you could take us a little bit more into the cell biology of this and tell us about beta-adrenergic activation and white cells, macrophages, either in the proinflammatory state and in the anti-inflammatory state and resolution state.

And then the second question was at least with respect to TNF-alpha, it's my understanding that TNF-alpha can be both pro- and anti-inflammatory depending on whether it's membrane-bound or whether it's released into the triad, and I wonder, if you're just measuring cytokine levels, are you missing subtle changes but very real changes in the processing of the cytokines?

Response From Marc Jeschke

Thank you very much, excellent question. Maybe I'll go with the first one, so the IL-6 I suspect is what you mean. There is, yes, both soluble and receptor-bound IL-6, and you're absolutely right. Actually, we looked recently in a paper in Shock, and there is a clear difference, but for this paper, we did not differentiate between the soluble and receptor-bound. This was just globally measuring IL-6, but you're absolutely right, there is so-called anti-inflammatory IL-6. For this one, we just can assume it is the pro-inflammatory IL-6.

And again, the variability, I'm not entirely sure, but maybe it is age-related, and we haven't studied this, but again, if you take a population from two years to 16, I think if you just maybe eliminate the very small kids, they are somewhat more homogeneous, and when we look really at an 18- to almost 60-year-olds, I think that there is a bigger variability in terms of the cytokines. That's why we see differences, but you're right, these subtle differences may be present, but we don't know.

And in terms of your receptor level and cell level, so I think the underlying mechanism—I didn't show the data because we showed in before in several reports that how browning occurs after burn, it is IL-6 driven, that is released from the bone marrow attracting macrophages via CCR2 migrating to the adipose tissue, and shift to a M2 phenotype, a more pro-inflammatory phenotype, releasing catecholamines, and that causes browning, lipolysis, and subsequent events. We looked recently, and it's actually a paper that's accepted in Annals as of last week, at that now more on a molecular level about beta-adrenergic receptors, and it seems to be a beta-2 phenomenon, but there's also beta-3 involvement. I suspect propranolol also has a very nonselective beta-3 effect, which then subsequently inhibits various responses on a tissue level, but it is very dominant signal that propranolol acts its main effective at the adipose tissue, and we have been ignoring it for a long time, and I think we as burn surgeons have to pay attention from a metabolic inflammatory aspect.

The unfortunate thing is that, as you know, lipids are very complex, and you have the same lipid there, and the metabolism is a very different pathway. One could be pro, one is anti, and it's so complex that it's going to be very difficult to dissect this out, but this is what we're doing right now to really understand more about the lipid metabolic response.

Thank you.

Dr. Mitchell Cohen (Denver, CO)

Mitch Cohen, Denver. Very beautiful work, and it builds on a beautiful body of work.

Response From Marc Jeschke

Thank you.

Dr. Mitchell Cohen (Denver, CO)

I'm curious about the outliers. I know that you weren't powered to see a mortality difference, and I'm intrigued by your last statement, which is that there's a metabolic association of the survival benefit, but I'm not sure it's shown here. There's a lot of heterogeneity in the metabolomic response or lipidomic and metabolomic response in the two groups. If you look at the subsegmented groups, patients who lived and died with propranolol, people that lived and died without, what do those metabolic differences look like? That's the first question.

And I guess the second question is I'm just interested because I don't know the literature, what happens in sympathectomized animals? Is there data to show that these mechanisms are the same? Have you looked at these same metabolomic or lipidomic mechanisms in a more controlled animal or rodent population to see what happens? I was very lucky at the very beginning of my career as a T32 fellow to work with Dr. Gamelli and look at sympathectomized animals and macrophage function. It seems like there should be a natural extension of these very same pathways now that we have much better technology to look at the transomic response of these animals.

Thanks for the work.

Response From Marc Jeschke

Thank you for your ents, so first, looking at the outliers, we tried to do this. It is just a very difficult aspect, as the sample size is small. What we're doing now is single-cell nuclear assays to look at that on exactly those patients because an n of 1 is actually kind of sufficient to get some ideas what is going on, but even for omics, we learned a very important lesson for omics in adults, which is you have to have at least five patients, otherwise you really are more into a speculation, and it's very difficult to conduct principal component analyses that are sufficient and redundant, so we did have a learning curve, and so now we are applying this learning going forward to all of our other data and looking particularly at the extremes. That's a good question.

And your question is actually quite brilliant about sympathectomized, so if you sympathectomize an animal, the animal doesn't do really well. We tried this. Dr. Wolf and I myself did this a long time ago. They don't do well. We tried to give other inhibitors, and they didn't really do well, so we couldn't really study this, but my brilliant postdoc who defended yesterday her PhD and is now a postdoc as of today, she did actually find acetylcholine is involved as a downstream signal from catecholamines, which is very intriguing because apparently acetylcholine signaling increases lipolysis and browning, so again, this is just preliminary data, but we are pursuing to get some better understanding about how the signaling occurs and what other downstream effects, but at this time we're looking into the depths of that.

Thank you.

Dr. Craig Coopersmith (Atlanta, GA)

Hi, Craig Coopersmith, Emory University. Great talk. A couple of questions. You said, getting back to our concept of religion, that you give propranolol to some patients and not to others. How do you make that determination at the bedside who you think is going to be helped and who is not, and does this work actually move you towards precision medicine about who you think might benefit and who might not?

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Response From Marc Jeschke

That's a great question. Thank you for your question. I give it to a patient where I really look at the resting energy expenditure when we do metabolic cart measurements and see, "Oh, my God, the resting energy expenditure is massively elevated," patient is persistently tachycardic despite adequate fluid, despite operation, despite nutrition, and the patient just looks extremely stressed. We don't unfortunately have an HPLC right there at the bedside to run catecholamines in the serum. As you know, they're very highly variable, and there were assays and attempts to do this, and even VMA in the urine is going to be difficult to extract, but based on that, and you can see a patient who is stressed, is hypermetabolic, and goes into the wrong direction if I start it, and again, yes, it is personalized medicine, and I think also what we are trying to do is actually take samples from the first operations and look at the trajectory of this patient. That's our goal or my goal over the next 10 years, the goal, exactly what you asked me, the approach of personalized medicine, what's the trajectory of this patient, where's the patient going, how can we shift this trajectory from a bad outcome to a good outcome, and maybe that technology will enable us to do so.

Dr. Nicole Gibran (Seattle, WA)

Nice presentation, Dr. Jeschke, and thank you for really helping us understand the science behind what propranolol might do at a cellular level. I'm less concerned about the effect on mortality because I think that as you've shown, there's opportunity to improve recovery for survivors. I do want to follow up on the issue of heterogeneic responses. I don't see that you've looked much at the possibility of polymorphisms in the beta-adrenergic receptors. What do you know about those genetic changes in your subjects?

Response From Marc Jeschke

That's an excellent question, Dr. Gibran, thank you. We did not look at polymorphism, but we know from the literature it's about 8% to 10% you have a genetic polymorphism where you have a beta-blockade resistance, so to speak, and you don't have this effect. We looked at our patients, and the study protocol—I didn't go into details, but the protocol to start 10 mg and go up based on your heart rate at 6 AM to come under 100 bpm, so none of our patients did not respond so I maybe can assume there was no resistance to beta antagonistic treatment, but we didn't test for this, so it's a good question, and it's maybe a flaw we can add to our discussions, and yes, we really don't know, but again, heterogeneity can answer, but the patients did all respond, and therefore it was not entirely a blinded study because the nurses at the bedside clearly saw the patients heart rate dropping, hence he's responding to propranolol.

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